Ex. No: 06	INTEGRATE ANIMATIONS AND TRANSITION
Date of Exercise	08.09.2026
Git-hub Host Link	

Aim:

To Implement animations and smooth transitions in a Flutter application using AnimatedContainer.

Description:

This experiment demonstrates the use of Flutter's **AnimatedContainer** to create a smooth size transition. A button is provided to change the size of the widget dynamically using setState(). A simple **light-purple color theme** is used for the user interface.

Procedure:

Step 1: Create a new Flutter project using VS Code or the terminal.

flutter create animation_app

cd animation_app

code .

Step 2: Open the lib/main.dart file and remove the default Flutter code.

Step 3: Create a StatefulWidget to manage the animation state.

Step 4: Create a Boolean state variable isLarge to control the size of the animated widget.

Step 5: Add an AnimatedContainer with a duration of one second and set its width and height based on the isLarge value.

Step 6: Add an **Animate** button and use setState() to change the value of isLarge.

Step 7: Apply a light purple background, app bar, animated container, and button to create a simple user interface.

Step 8: Save the program and run the application using:

flutter run -d chrome

Step 9: Click the **Animate** button and observe the smooth change in the size of the widget.

Program:

```
import 'package:flutter/material.dart';
void main() {
  runApp(const AnimationApp());
}
class AnimationApp extends StatelessWidget
  { const AnimationApp({super.key});

  @override
  Widget build(BuildContext context)
    { return
    MaterialApp(  debugShowCheckedMode
    Banner: false, title: 'Flutter Animation',
    theme: ThemeData(
        colorScheme:
          ColorScheme.fromSeed( seedColor:
          Colors.deepPurple,
        ),
        useMaterial3: true,
      ),
      home: const AnimationPage(),
    );
  }
}
class AnimationPage extends StatefulWidget
  { const AnimationPage({super.key});

  @override
  State<AnimationPage> createState() => _AnimationPageState();
}
class _AnimationPageState extends State<AnimationPage>
  { bool isLarge = false;
  @override
  Widget build(BuildContext context)
    { return Scaffold(
      appBar:
        AppBar( title:
        const Text(
          'Flutter Animations',
          style:
          TextStyle( color:
          Colors.white,
            fontWeight: FontWeight.bold,
          ),
        ),
```

```
    centerTitle: true,
    backgroundColor: Colors.deepPurple,
  ),
  body:
    Container( width:
    double.infinity,
    height: double.infinity,
    color: const Color(0xFFEDE7F6),

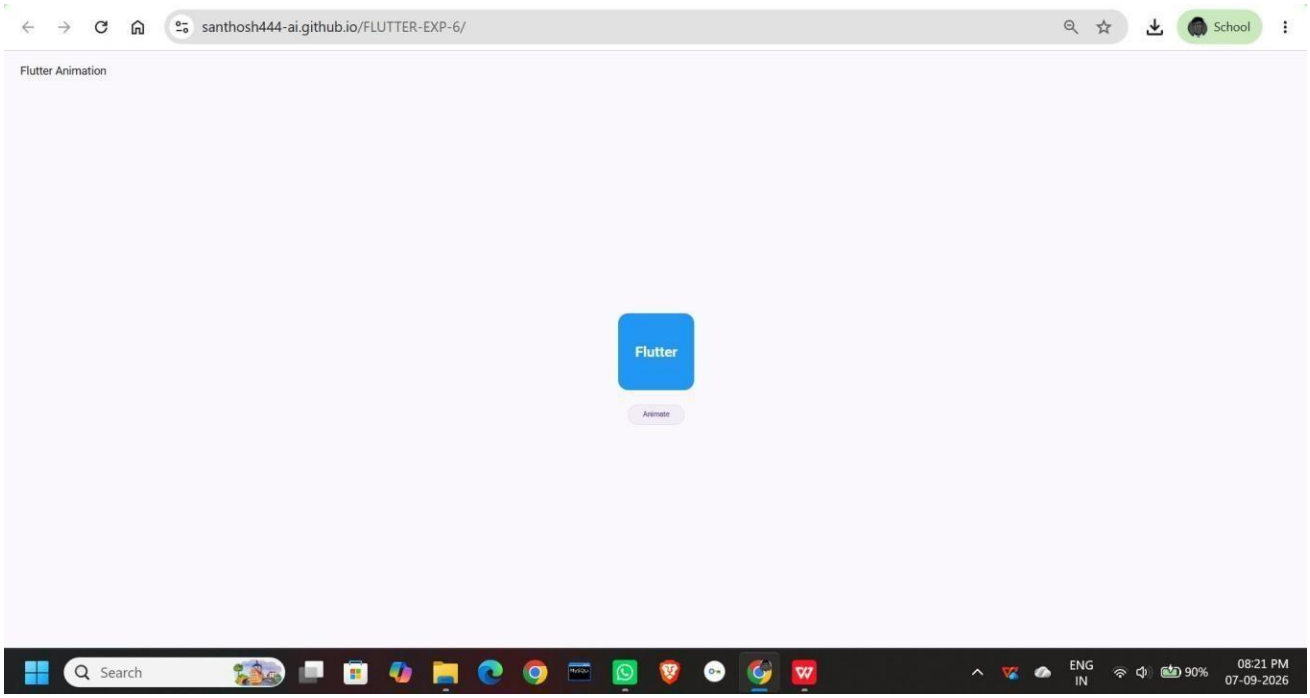
    child:
      Center( child:
      Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          AnimatedContainer(
            duration: const Duration(seconds: 1),
            curve: Curves.easeInOut,
            width: isLarge ? 300 : 150,
            height: isLarge ? 300 : 150,
            decoration: BoxDecoration(
              color: const Color(0xFFB39DDB),
              borderRadius:
              BorderRadius.circular( isLarge ? 50 :
              20,
              ),
              boxShadow: const
                [ BoxShadow( blurRadi
                us: 15,
                  spreadRadius: 3,
                  offset: Offset(0, 8),
                ),
              ],
            ),
            child: const
              Center( child:
              Text( 'Flutter',
                style:
                  TextStyle( font
                  Size: 28,
                  color: Colors.white,
                  fontWeight: FontWeight.bold,
                ),
              ),
            ),
          ),
          const SizedBox(height: 40),
```

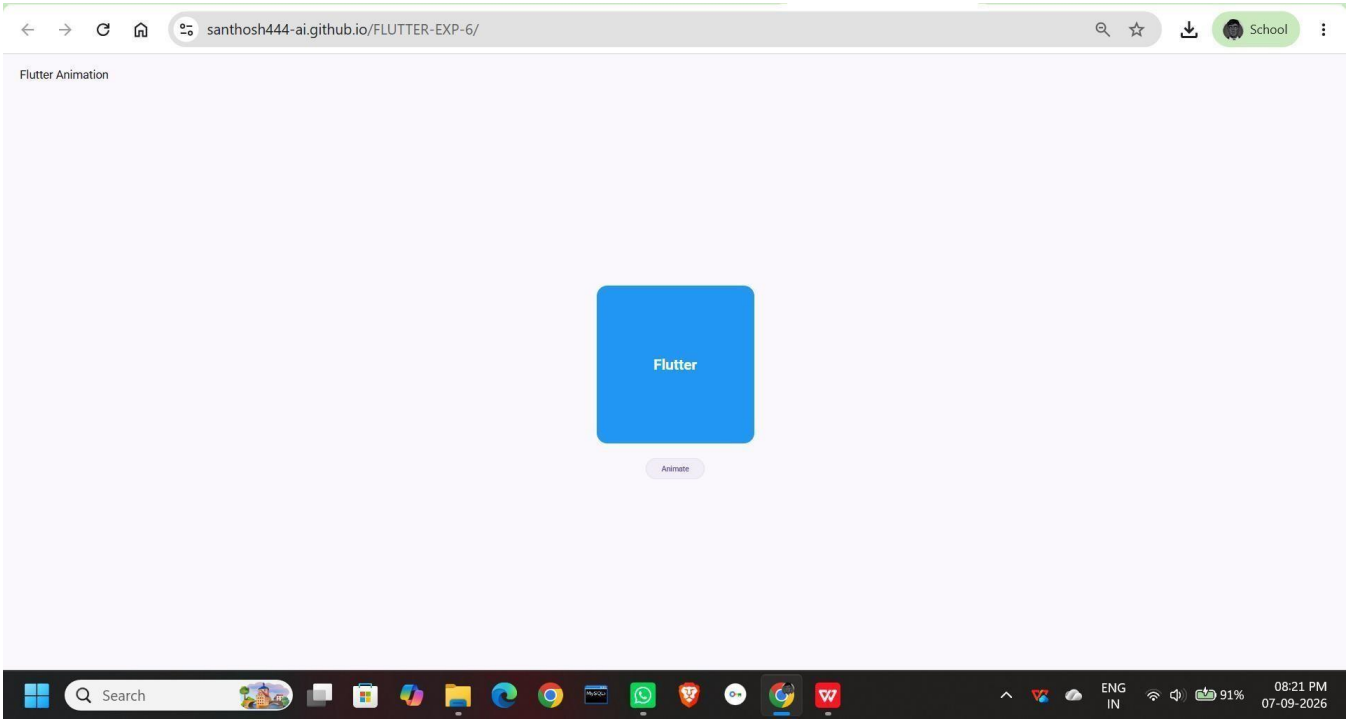
```
ElevatedButton.icon( onPres
sed: () {
    setState(() {
      isLarge = !isLarge;
    });
  },
  icon: const Icon(Icons.animation),
  label: const Text(
    'Animate',
    style:
      TextStyle( font
      Size: 18,
      fontWeight: FontWeight.bold,
    ),
  ),
  style:
    ElevatedButton.styleFrom( background
    Color: Colors.deepPurple,
    foregroundColor: Colors.white,
    padding: const
    EdgeInsets.symmetric( horizontal: 35,
      vertical: 16,
    ),
    shape:
      RoundedRectangleBorder( borderRadius:
      BorderRadius.circular(30),
    ),
  ),
),

const SizedBox(height: 20),

Text(
  isLarge ? 'Widget is Large' : 'Widget is Small',
  style: const TextStyle(
    color: Colors.deepPurple,
    fontSize: 18,
    fontWeight: FontWeight.w500,
     ),),],),),),);}}
```

Output:


santhosh444-ai.github.io/FLUTTER-EXP-6/
School
Flutter Animation
Flutter
Animate
Search
ENG
IN
90%
08:21 PM
07-09-2026



Result:

The Flutter application was successfully implemented using **AnimatedContainer**. The widget smoothly changes its size when the **Animate** button is clicked, demonstrating basic animation and transition functionality in Flutter.